1. $MAPbI_3$

Table 2. Band Gap, Average Absolute NA Coupling, Pure-Dephasing Time, Nonradiative Electron–Hole Recombination Time, and Radiative Lifetime for the VBM–CBM Transition in the Ortho-162 K, Tetra-162 K, Tetra-315 K, and Cubic-315 K $MAPbI_3$ Systems

	band gap (eV)	NA coupling (meV)	dephasing (fs)	nonradiative recombination (ns)	radiative recombination (ns)
Ortho-162 K	1.65[44]	1.33	24.7	1.19	1.47
Tetra-162 K	1.68[131]	1.10	18.2	1.34	2.23
Tetra-315 K	1.68[131]	0.93	14.1	2.47	3.03
Cubic-315 K	1.55[132]	0.74	7.8	5.62	7.22

Ref. J. Am. Chem. Soc. 2022, 144, 19137-19149
https://doi.org/10.1021/jacs.2c08627

2. $MAPbI_3$ (including nuclear quantum effects)

Table 2. Canonically Averaged Band Gap, Absolute NAC, Pure-Dephasing Time, and Nonradiative and Radiative Recombination Times for the VBM–CBM Transition of $MAPbI_3$ Obtained Using AIMD and RPMD at 160 and 330 K

		bandgap (eV)	NAC (meV)	dephasing (fs)	nonradiative recombination (ns)	radiative recombination (ns)
160 K	AIMD	1.71	1.51	11.57	0.95	2.57
	RPMD	1.61	1.02	6.63	2.57	3.57
330 K	AIMD	1.72	1.22	8.34	2.16	2.83
	RPMD	1.69	0.96	5.88	2.92	3.99

Ref. J. Am. Chem. Soc. 2023, 145, 14112−14123
https://doi.org/10.1021/jacs.3c04412

3. $MAPbBr_3$

Table 2. Energy Gap, Averaged Absolute Value of NA Coupling, Pure-Dephasing Time, and Electron Trapping and Recombination Times in $MAPbBr_3$, Br_i, Br_i^-, and Br_i^+ Systems

		energy gap (eV)	NA coupling (meV)	dephasing (fs)	time (ns)
$MAPbBr_3$	CBM → VBM	2.34	1.90	14.9	1.16
Br_i	CBM → VBM	2.64	1.31	9.84	3.53
Br_i^-	CBM → VBM	2.40	1.79	13.5	1.17
	VBM → trap	1.16	2.87	1.80	0.26
Br_i^+	trap → CBM	1.23	2.53	1.79	0.81
	CBM → VBM	2.39	2.20	7.06	0.84

Ref. J. Phys. Chem. Lett. 2022, 13, 4193−4199
https://pubs.acs.org/doi/10.1021/acs.jpclett.2c00965

4. $FAPbI_3$

Table 1. Canonically Averaged Bandgap, Average Absolute NA Coupling, Initial Value of the Un-ACF, Pure-Dephasing Time, and Nonradiative Electron–Hole Recombination Time for Pristine $FAPbI_3$ at 300 K and $FAPbI_3$ Containing Σ3 (111) TB at 300 and 400 K

system	bandgap (eV)	NA coupling (meV)	un-ACF initial value (eV^2)	pure-dephasing (fs)	recombination (ns)
pristine-300 K	1.47	0.19	0.03325	5.57	7.19
TB-300 K	1.58	0.15	0.06048	2.62	16.8
TB-400 K	1.59	0.13	0.07325	2.09	19.9

Ref. J. Phys. Chem. Lett. 2021, 12, 8575−8582
https://doi.org/10.1021/acs.jpclett.1c02653

5. $CsPbI_3$

Table 2. Experimental Bandgap, Average Absolute NA Coupling, Pure-Dephasing Time, and Nonradiative Electron–Hole Recombination Time for $CsPbI_3$, $CsPbI_2Br$, and $CsPbI_{1.88}BrAc_{0.12}$

system	bandgap (eV)	NA coupling (meV)	pure-dephasing (fs)	recombination (ns)
$CsPbI_3$	1.73[93]	1.29	10.26	2.16
$CsPbI_2Br$	1.92[24]	0.93	12.40	12.71
$CsPbI_{1.88}BrAc_{0.12}$	1.96[33]	0.67	15.59	31.71

Ref. J. Phys. Chem. Lett. 2022, 13, 11375−11382
https://doi.org/10.1021/acs.jpclett.2c03072

6. $CsPbBr_3$

Table 2. Average Bandgap (eV), Averaged Absolute NA Coupling (meV), Dephasing Time (fs), and Nonradiative Electron–Hole Recombination Time Scale (ps) Obtained Using the Normal and Frozen MD Trajectories[a]

	bandgap (eV)	NA coupling (meV)	dephasing time (fs)	recombination (ps)
normal MD	2.35	1.44	6.2	862
forzen MD	2.34	1.68	6.7	632

Ref. J. Phys. Chem. Lett. 2022, 13, 3016−3022
https://doi.org/10.1021/acs.jpclett.2c00532